

Project Initialization and Planning Phase

Date	01.10.2026
Team ID	SWTID-2026-1380
Project Title	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report presents PocketSmart AI, a GenAI-powered budget and recommendation assistant designed to help users plan purchases and activities within a fixed budget. It provides personalized recommendations for home interior planning, party/event planning and jewelry selection, with optional image-based jewelry recommendations.

Project Overview	
Objective	The primary objective is to provide budget-aware, personalized recommendations that help users make faster and more informed decisions for home interiors, parties/events and jewelry.
Scope	The system covers a responsive web interface, user authentication, three planning modules, AI-based recommendation generation, optional jewelry image input, external product/service links and recommendation history.
Problem Statement	
Description	Users often have difficulty comparing many products and services while staying within a fixed budget. Manual searching across multiple platforms is time-consuming and can make decision-making difficult.
Impact	A budget-aware recommendation assistant can reduce manual comparison, save time and present relevant options based on the user's requirements and available budget.
Proposed Solution	
Approach	PocketSmart AI collects the user's budget and requirements, validates the input, sends a structured request to the recommendation service, and presents suitable recommendations. Gemini is used for AI-generated recommendations, with fallback/default recommendations when the AI service is unavailable.
Key Features	• Home Interior Planning • Party/Event Budget Planning • Jewelry Recommendations • Optional jewelry outfit-image input • Budget-aware AI recommendations • User authentication and recommendation history

	• Personalized recommendation results • Relevant external product/service links • Responsive web interface
--	--

Resource Requirements

Resource Type	Description	Specification/Allocation
Standard laptop/PC; internet connection		
Computing Resources	CPU/GPU specifications	Standard CPU; dedicated GPU is not required for the application because Gemini is accessed through an API
Memory	RAM specifications	8 GB RAM or above recommended
Storage	Disk space for application, database and uploaded files	At least 10 GB free space for development
Software		
Frameworks	Python framework	FastAPI
Libraries	Additional libraries	SQLAlchemy, Pydantic Settings, Jinja2, Werkzeug, Pillow, google-genai
Development Environment	IDE / tools	VS Code or PyCharm; Python virtual environment
User accounts, planner inputs and recommendation history stored in SQLite; optional jewelry images are processed for recommendations		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv